

Analysed candidate hiring data to optimize recruitment performance

SQL Phase

Export Cleaned Data from Python

```
df.to_excel("Recruitment_Cleaned.xlsx", index=False)
```

Import Cleaned Excel data file into SQL

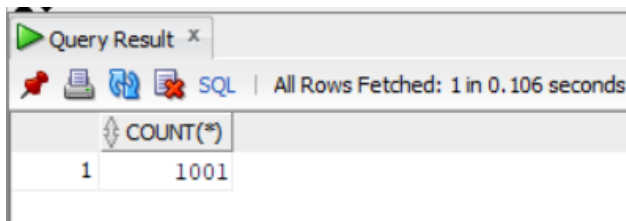
Oracle SQL Developer → Import Data Wizard

After import, verify: `SELECT * FROM RECRUITMENT;`

Data Validation

Total Records :

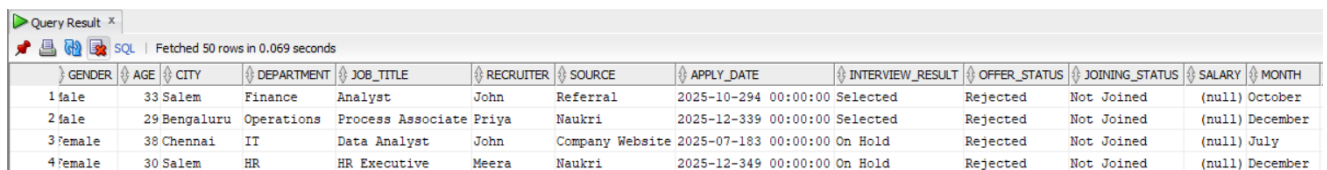
```
SELECT COUNT(*) FROM RECRUITMENT;
```



COUNT(*)
1001

Check Missing Salary :

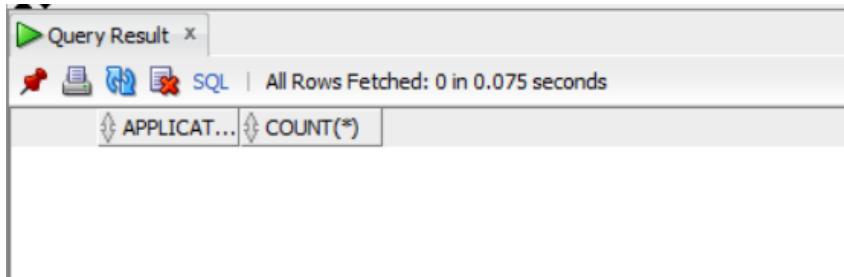
```
SELECT *  
FROM RECRUITMENT  
WHERE SALARY IS NULL;
```



GENDER	AGE	CITY	DEPARTMENT	JOB_TITLE	RECRUITER	SOURCE	APPLY_DATE	INTERVIEW_RESULT	OFFER_STATUS	JOINING_STATUS	SALARY	MONTH
1	female	33 Salem	Finance	Analyst	John	Referral	2025-10-294 00:00:00	Selected	Rejected	Not Joined	(null)	October
2	female	29 Bengaluru	Operations	Process Associate	Priya	Naukri	2025-12-339 00:00:00	Selected	Rejected	Not Joined	(null)	December
3	female	38 Chennai	IT	Data Analyst	John	Company Website	2025-07-183 00:00:00	On Hold	Rejected	Not Joined	(null)	July
4	female	30 Salem	HR	HR Executive	Meera	Naukri	2025-12-349 00:00:00	On Hold	Rejected	Not Joined	(null)	December

Check Duplicate Applications :

```
SELECT APPLICATION_ID, COUNT(*) FROM RECRUITMENT  
GROUP BY APPLICATION_ID HAVING COUNT(*) > 1;
```

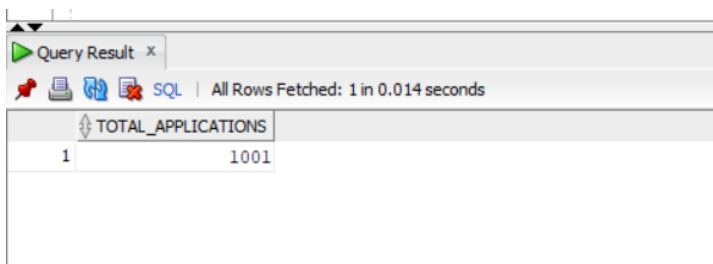


APPLICATION_ID	COUNT(*)
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Basic SQL Analysis

Total Applications :

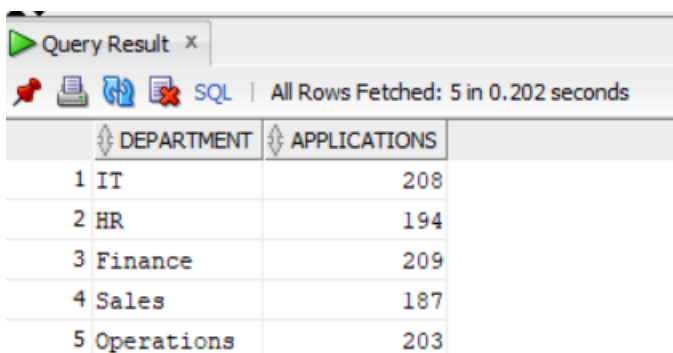
```
SELECT COUNT(*) AS TOTAL_APPLICATIONS FROM  
RECRUITMENT;
```



TOTAL_APPLICATIONS
1001

Department-wise Applications :

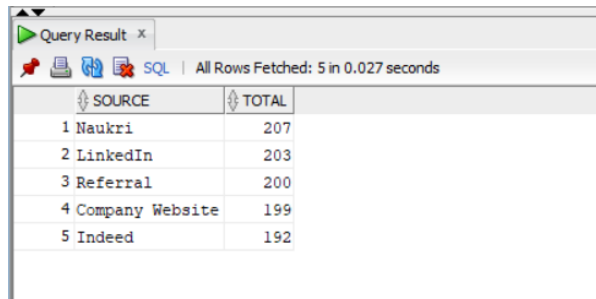
```
SELECT DEPARTMENT, COUNT(*) AS APPLICATIONS FROM  
RECRUITMENT GROUP BY DEPARTMENT;
```



DEPARTMENT	APPLICATIONS
1 IT	208
2 HR	194
3 Finance	209
4 Sales	187
5 Operations	203

Recruitment Source Analysis:

```
SELECT SOURCE, COUNT(*) AS TOTAL FROM RECRUITMENT  
GROUP BY SOURCE ORDER BY TOTAL DESC;
```



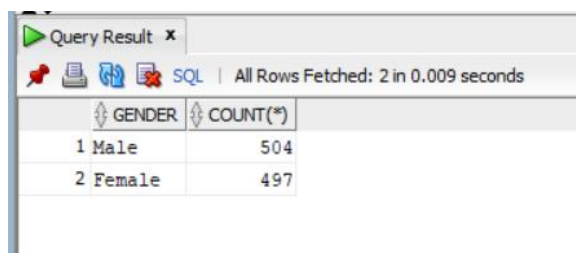
Query Result x

SQL | All Rows Fetched: 5 in 0.027 seconds

SOURCE	TOTAL
1 Naukri	207
2 LinkedIn	203
3 Referral	200
4 Company Website	199
5 Indeed	192

Gender Distribution :

```
SELECT GENDER, COUNT(*) FROM RECRUITMENT GROUP BY  
GENDER;
```



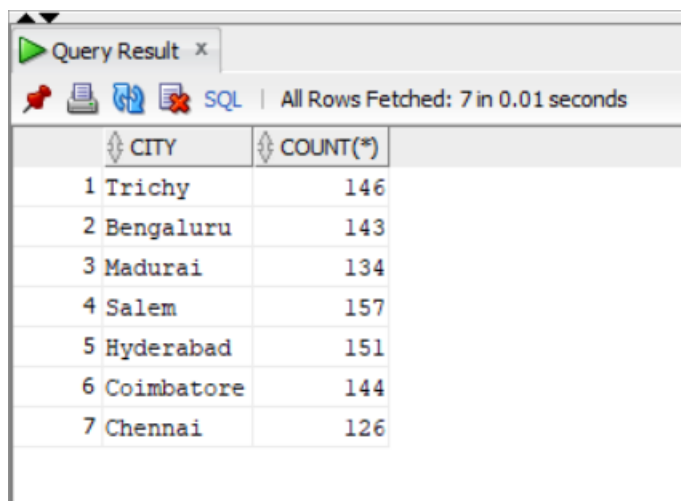
Query Result x

SQL | All Rows Fetched: 2 in 0.009 seconds

GENDER	COUNT(*)
1 Male	504
2 Female	497

City-wise Candidates:

```
SELECT CITY, COUNT(*) FROM RECRUITMENT GROUP BY CITY;
```



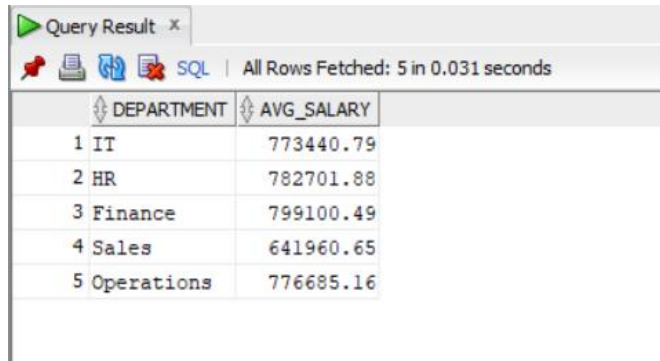
Query Result x

SQL | All Rows Fetched: 7 in 0.01 seconds

CITY	COUNT(*)
1 Trichy	146
2 Bengaluru	143
3 Madurai	134
4 Salem	157
5 Hyderabad	151
6 Coimbatore	144
7 Chennai	126

Average Salary by Department:

```
SELECT DEPARTMENT, ROUND(AVG(SALARY),2) AS  
AVG_SALARY FROM RECRUITMENT GROUP BY DEPARTMENT;
```



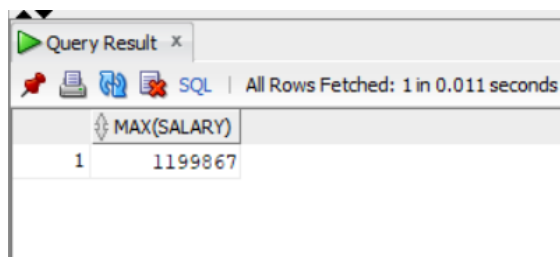
Query Result x

All Rows Fetched: 5 in 0.031 seconds

	DEPARTMENT	AVG_SALARY
1	IT	773440.79
2	HR	782701.88
3	Finance	799100.49
4	Sales	641960.65
5	Operations	776685.16

Highest Salary:

```
SELECT MAX(SALARY) FROM RECRUITMENT;
```



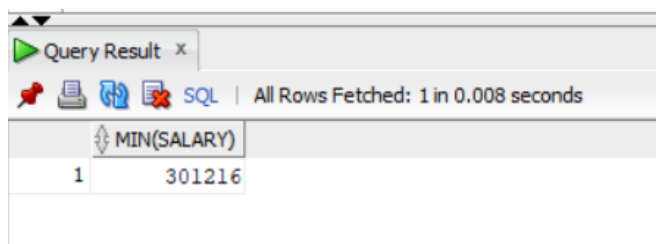
Query Result x

All Rows Fetched: 1 in 0.011 seconds

	MAX(SALARY)
1	1199867

Lowest Salary:

```
SELECT MIN(SALARY) FROM RECRUITMENT;
```



Query Result x

All Rows Fetched: 1 in 0.008 seconds

	MIN(SALARY)
1	301216

Top 10 Highest Salaries:

```
SELECT * FROM (SELECT * FROM RECRUITMENT WHERE  
SALARY IS NOT NULL ORDER BY SALARY DESC) WHERE  
ROWNUM <= 10;
```

Query Result

All Rows Fetched: 10 in 0.024 seconds

GENDER	AGE	CITY	DEPARTMENT	JOB_TITLE	RECRUITER	SOURCE	APPLY_DATE	INTERVIEW_RESULT	OFFER_STATUS	JOINING_STATUS	SALARY	MONTH
1male	34	Trichy	IT	Data Analyst	Vikram	Indeed	2025-10-292 00:00:00	Selected	Accepted	Joined	1159667	October
2male	29	Hyderabad	IT	Data Analyst	Vikram	Company Website	2025-05-130 00:00:00	Selected	Accepted	Pending	1198389	May
3ile	34	Hyderabad	IT	Data Analyst	John	Company Website	2025-03-77 00:00:00	Selected	Pending	Pending	1154967	March
4ile	25	Salem	IT	Data Analyst	Meera	Company Website	2025-02-36 00:00:00	Selected	Accepted	Pending	1189337	February
5male	33	Madurai	Finance	Analyst	Vikram	Naukri	2025-10-298 00:00:00	Selected	Pending	Pending	1185649	October
6ile	21	Salem	Operations	Process Associate	Suresh	Referral	2025-12-345 00:00:00	Selected	Pending	Pending	1183947	December
7male	29	Bengaluru	Operations	Process Associate	John	Company Website	2025-02-51 00:00:00	Selected	Accepted	Joined	1178221	January
8male	22	Madurai	Finance	Analyst	John	Indeed	2025-06-158 00:00:00	Selected	Accepted	Joined	1177609	June
9male	29	Salem	IT	Data Analyst	Vikram	Referral	2025-02-59 00:00:00	Selected	Pending	Pending	1173131	February
10male	29	Bengaluru	Operations	Process Associate	Vikram	Naukri	2025-04-107 00:00:00	Selected	Pending	Pending	1170240	April

Intermediate SQL :

```
SELECT DEPARTMENT, COUNT(*) TOTAL_HIRES, AVG(SALARY)
AVG_SALARY FROM RECRUITMENT WHERE
JOINING_STATUS='Joined' GROUP BY DEPARTMENT HAVING
COUNT(*) > 10;
```

Query Result

All Rows Fetched: 5 in 0.006 seconds

DEPARTMENT	TOTAL_HIRES	AVG_SALARY
1 IT	13	753696.2307692307692307692307692308
2 HR	15	763705.33333333333333333333333333333333
3 Finance	18	874065.05555555555555555555555555555556
4 Sales	11	662039.27272727272727272727272727272727
5 Operations	12	824084.75

Window Functions:

```
SELECT CANDIDATE_NAME, DEPARTMENT, SALARY, RANK()  
OVER(ORDER BY SALARY DESC) SALARY_RANK FROM  
RECRUITMENT;
```

Query Result x

SQL | Fetched 50 rows in 0.027 seconds

	CANDIDATE_NAME	DEPARTMENT	SALARY	SALARY_RANK
1	Sneha S	Finance	(null)	1
2	Vijay K	Operations	(null)	1
3	Sneha R	IT	(null)	1
4	Priya R	HR	(null)	1
5	Vijay M	Sales	(null)	1
6	Vijay K	Operations	(null)	1
7	Meena R	HR	(null)	1

Department-wise Salary Rank:

```
SELECT CANDIDATE_NAME, DEPARTMENT, SALARY,
DENSE_RANK() OVER(PARTITION BY DEPARTMENT ORDER BY
SALARY DESC ) DEPT_RANK FROM RECRUITMENT;
```

Query Result x			
SQL Fetched 50 rows in 0.013 seconds			
	CANDIDATE_NAME	DEPARTMENT	SALARY
1	Meena S	Finance	(null)
2	Sneha R	Finance	(null)
3	Anita S	Finance	(null)
4	Arun K	Finance	(null)
5	Ravi M	Finance	(null)

Running Total Salary:

```
SELECT APPLICATION_ID, SALARY, SUM(SALARY)
OVER(ORDER BY APPLICATION_ID) FROM RECRUITMENT;
```

Query Result x			
SQL Fetched 50 rows in 0.013 seconds			
	APPLICATION_ID	SALARY	SUM(SALARY)OVER(ORDERBYAPPLICATION_ID)
1	A0001	(null)	(null)
2	A0002	412774	412774
3	A0003	(null)	412774
4	A0004	(null)	412774
5	A0005	(null)	412774

Create SQL Views

View 1

```
CREATE VIEW VW_RECRUITMENT_SUMMARY AS SELECT
DEPARTMENT, COUNT(*) TOTAL_APPLICATIONS, AVG(SALARY)
AVG_SALARY, COUNT(CASE WHEN JOINING_STATUS='Joined' THEN 1
END) TOTAL_HIRES FROM RECRUITMENT GROUP BY DEPARTMENT;
```

View 2

```
CREATE VIEW VW_SOURCE_ANALYSIS AS SELECT SOURCE, COUNT(*)
APPLICATIONS FROM RECRUITMENT GROUP BY SOURCE;
```

View 3

```
CREATE VIEW VW_RECRUITER_PERFORMANCE AS SELECT
RECRUITER, COUNT(*) APPLICATIONS FROM RECRUITMENT GROUP
BY RECRUITER;
```